Free Speech and the Fake News Problem[1]

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Abstract:

How should a platform or a society address the spread of misinformation? Scholars often repeat, and courts often assume, that the best test of truth is the power to get itself accepted in the competition of the market. While such writing focuses on veracity, this Essay proposes that harms are better cognized as decision errors and externalities. A focus on truth per se is misguided because one cannot own truth and one cannot be dispossessed of truth. By contrast, one can be liable for decision errors and externalities. This alternate interpretation suggests a syllogism that redefines the problem:

> *Misinformation produces externalities. Externalities produce market failures. Market failures require intervention, but the First Amendment forbids government intervention in speech. Thus, attempts by courts and society to push the problem to the "marketplace of ideas" to sort things out will fail. Markets do not self-correct market failures.*

This Essay then offers a message of hope. Using this new definition, it proposes to design a system of rights such that *decentralized* institutions can internalize negative externalities. Society might then curb misinformation with no censorship at all and no central authority – not government, not private firms, and not powerful individuals – judging truth. The task of addressing fake news is transformed into a task of problem definition, decentralized mechanism design, and truthful signaling for which tools of information economics are well-suited.

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This is the first of two essays addressing the challenge of false claims in free speech. It focuses on externalities. The second essay, *A Consequentialist Approach to Free Speech*, focuses on decision errors.

[1] The latest draft is available at http://ssrn.com/abstract=4414261 .

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I. Introduction

Fake news is a problem.[2] It is a near universal problem.[3] Trolls and propagandists have used it as a weapon in political campaigns, anti-vaccination campaigns, nutrition battles, insurrections, and to sow ethnic conflict. Social costs of fake news include inaccurate beliefs, worse democratic processes for selecting high-quality candidates, increased skepticism of legitimate sources, and reduced incentives to produce truthful news.[4] The inventor of the World Wide Web identified fake news as one of the most dangerous assaults on the Internet.[5] Fake news is not new. Stone frescoes record the victories of "Rameses the Great" on temples from the thirteenth century BC yet more complete records show his battle was a stalemate.[6]

Fake news is a known problem. Figure 1 shows occurrences of the terms "fake news," "false news," "misinformation," and "disinformation" in literature since 1800. Until recently, misinformation has been the more common term. Unsurprisingly, "misinformation" rose during both World War I and World War II as conflicting powers sought to demoralize each other's troops with concocted stories of false victories.[7] "Disinformation" spiked during the Cold War between the Soviet Union and the United States, then rose again under the administration of Vladimir Putin.[8] "Fake news" rose under the Trump administration. Deceit is common among adversaries and those pushing hidden agendas.

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[2] Tim Wu, *Is the First Amendment Obsolete?* 117 MICH. L. REV. 547 (2018); Richard Hasen, *Cheap Speech and What It Has Done to American Democracy*, 200 FIRST AMEND. L. REV. 16 (2017); YOCHAI BENKLER ET AL., NETWORK PROPAGANDA (2018).

[3] The Wikipedia entry on fake news lists accounts of problems across all seven continents. *See Fake News*, WIKIPEDIA https://en.wikipedia.org/wiki/Fake_news (last visited Sept. 26, 2022) (detailing accounts of fake news in over forty countries).

[4] Hunt Allcott & Matthew Gentzkow, *Social Media and Fake News in the 2016 Election*, 31 J. ECON. PERSP. 211, 213 (2017).

[5] *See* Jon Swartz, *The World Wide Web's Inventor Warns It's in Peril on 28th Anniversary*, USA TODAY (Mar. 11, 2017), https://eu.usatoday.com/story/tech/news/2017/03/11/world-wide-webs-inventor-warns-s-peril/99005906/.

[6] WILLIAM WEIR, HISTORY'S GREATEST LIES: THE STARTLING TRUTHS BEHIND WORLD EVENTS OUR HISTORY BOOKS GOT WRONG (2009).

[7] See Becky Little, *Inside America's Shocking WWII Propaganda Machine*, NAT'L GEOGRAPHIC (Dec. 19, 2016), http://news.nationalgeographic.com/2016/12/world-war-2-propaganda-history-books/.

[8] *See Disinformation*, WIKIPEDIA, https://en.wikipedia.org/wiki/Disinformation (last visited Sept. 23, 2022).

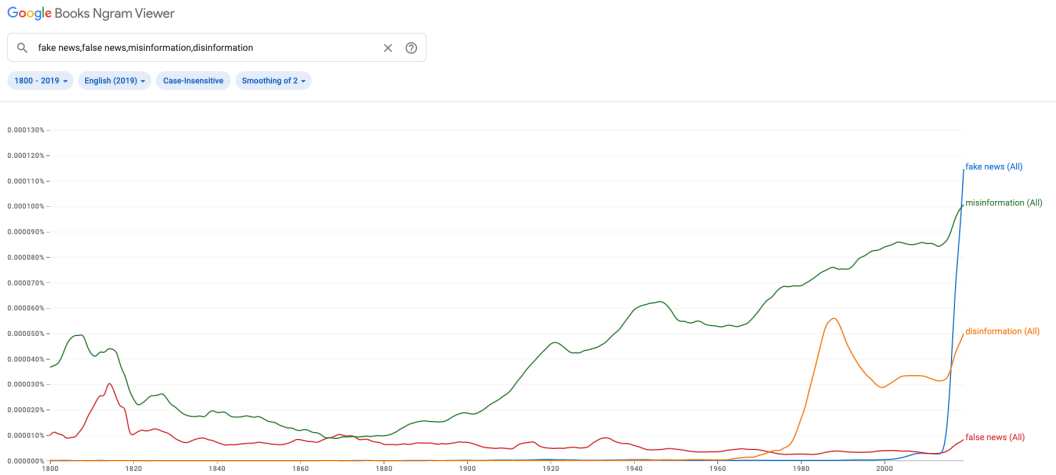

Figure 1 – Occurrence of fake news terminology in books and literature 1800-2019. Other terms dominated until 2016. A rise in misinformation occurred around World Wars I and II, a rise in disinformation occurred during the Cold War, and in fake news in 2016.[9]

Challenges of modern fake news, however, render it more pernicious than in the past. First, modern platforms allow secret or insulated public messaging. The propagandist can whisper his case at a scale that is simultaneously vast yet almost invisible to those who, upon observing the message, would oppose it with countervailing evidence. During the 2016 U.S. presidential election, members of one party could buy ads targeted at individual coal miners in Pennsylvania.[10] The timing and content of such messages could remain hidden from the opposing party in a manner that is largely impossible using traditional broadcast media.[11] Second, unlike traditional media that hired journalists and broadcasters, modern platforms do not create the content they distribute. Both by law and by design, they absolve themselves of responsibility for propagating fake news, even as they expand the population capable of spreading it. Broadening inclusivity cuts two ways. Modern platforms give voice to the human rights worker, the disenfranchised, the oppressed, the builder, and the whistleblower. They also give voice to the troll, the racist, the enemy state, the bot, and the bot army. Third, "deep fake" technology has created false realities.[12] Victims of political slander could once credibly deny as hearsay false and defamatory claims by third parties. Deep fake technology, however, can create first person fictions that make a victim appear to have committed acts he or she never did or spoken words he or she never would. In appearance and behavior, forged details have become indistinguishable from originals.

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[9] GOOGLE BOOKS NGRAM VIEWER, http://books.google.com/ngrams (select years "1800-2019", case-insensitive, smoothing 2, search "fake news, false news, misinformation, disinformation").

[10] The Trump campaign introduced disappearing ads on Facebook in Wisconsin, North Carolina and Georgia to deter black voters from casting ballots. *See* Channel 4 News, *Revealed: Trump Campaign Strategy to Deter Millions of Black Americans from Voting in 2016*, YOUTUBE (Sep. 28, 2020), https://www.youtube.com/watch?v=KIf5ELaOjOk.

[11] The counterargument that Clinton could have bought similar ads is too simplistic. Without knowing what has been said to whom, the cost of covering all possible arguments among all possible listeners is prohibitive. The chief beneficiary, in this case, would be the platform selling blanket advertising.

[12] Bobby Chesney & Danielle Citron, *Deep Fakes: A Looming Challenge for Privacy, Democracy, and National Security*, 107 CAL. L. REV. 1753, 1769 (2019).

Fourth, the scale of misinformation creates potential for systemic failure as social systems can exhibit cascade failures when pushed too far.[13] Like coral ecosystems that have withstood millennia yet passed in an ecological instant, social systems can break down when pressed beyond their institutional constraints and citizens cannot agree on basic facts.

Fake news is a hard problem.[14] Open societies tend toward higher growth rates,[15] greater freedom of expression, and greater justice[16] than closed societies. Censorship is a preferred tool of despots and dictators. Freedom of speech is a fundamental right precisely because it exposes those with much to hide. At the same time, a right to free speech must balance a right to privacy, a right to self-rule, and a right to be free of harms. Not every fact of one's private life deserves public scrutiny. No lie justifies overturning a fair election. Yet, it can also happen that ugly truths can be more divisive than harmless lies. Design of a fair and balanced news mechanism must weigh rights and properties that conflict with one another yet promote the public good. Presumably, the balance of these rights should be robust to circumstance and not the fragile reactions to politics of the times.

This Essay undertakes two tasks. The first is to articulate why fake news is a problem and how harm occurs. While much scholarship focuses on veracity, harms are better cognized as decision errors and externalities. A focus on truth per se is misguided because one cannot own truth and one cannot be dispossessed of truth. By contrast, one can be liable for decision errors and externalities. From this understanding, the second task is to articulate a new system of rights such that decentralized choices of speakers and listeners in the market for ideas, like those of suppliers and buyers in the market for products, lead to a more desirable social outcome. This removes the center, which limits the ability of any specific control point to censor the flow of speech. This task will also require the articulation of tests by which one speech intervention mechanism might be compared with another in order to rank more desirable outcomes. There is no promise that such a framework is either correct or complete, only that it is as balanced as this author can make it.

II. What is Fake News?

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[13] Joseph B. Bak-Coleman et al., *Stewardship of Global Collective Behavior*, 118 PROCEEDINGS OF THE NAT'L ACAD. OF SCI. OF THE U.S. 1 (2021).

[14] Hasen, *supra* note 2, at 230 ("[D]emocracy-related problems caused by the rise of cheap speech are not easily solvable."); BENKLER *supra* note 2 at 352 ("While the options are many, there are no silver bullets."); David Barker & Morgan Marietta, *Fact-Checking Can't Do Much When People's "Dueling Facts" are Driven by Values Instead of Knowledge*, NIEMAN LAB (May 8, 2019) https://www.niemanlab.org/2019/05/fact-checking-cant-do-much-when-peoples-dueling-facts-are-driven-by-values-instead-of-knowledge/ ("Perhaps the most disappointing finding from our studies . . . is that there are no known fixes to this problem.").

[15] Adam Przeworski et al., *Political Regimes and Economic Growth*, *in* DEMOCRACY AND DEVELOPMENT: PROCEEDINGS OF THE IEA CONFERENCE HELD IN BARCELONA, SPAIN 3, 3-27, (Amiya Kumar Bagchi ed. 1995).

[16] Daron Acemoglu et al., *Institutions as a Fundamental Cause of Long-Run Growth*, 1 HANDBOOK OF ECON. GROWTH 386, 386-472 (2005).

A. Definitions

To solve a problem requires that we first define it.[17] As different definitions of 'fake news' suggest different interventions, a brief summary of scholarship follows:

- Fabricated information that mimics news media content in form but not in organizational process or intent.[18]
- Fraudulent high velocity content[19] … or fraudulent news format.[20]
- News that is intentionally and verifiably false and could mislead readers.[21] Presentation of facts that is misleading by design.[22]
- 'Disinformation' is preferred to 'fake news' and is "…all forms of false, inaccurate, or misleading information designed, presented, and promoted to intentionally cause public harm or for profit."[23]
- Viewed narrowly, it is "verifiably false information" but, viewed broadly, it includes deliberate attempts at disinformation and news distortion for earning money, promoting ideology or confusion or polarization or sowing discontent.[24]
- Fake news is disseminated with the intent to mislead in order to damage an agency, entity, or person, and/or gain financially or politically,[25] often using sensationalist, dishonest, or outright fabricated headlines to increase readership, online sharing, and revenue.[26]
- News from any of three distinct categories: i) stories invented to make money or discredit others, ii) news based on fact but spun to suit an agenda, iii) news that people disagree with.[27]

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[17] "The topics of mis- mal- and dis- information are too important to start … regulating around until we have a shared understanding of what we mean by these terms." Clair Wardle & Hossein Derakshan, *Information Disorder: Toward an Interdisciplinary Framework for Research and Policymaking*, COUNCIL OF EUR. (Sept. 27, 2017), https://tverezo.info/wp-content/uploads/2017/11/PREMS-162317-GBR-2018-Report-desinformation-A4-BAT.pdf.

[18] David M. J. Lazer et al., *The Science of Fake News*, 359 SCI. 1094, 1094-1096 (2018).

[19] NICOLE STREMLAU ET AL., WORLD TRENDS IN FREEDOM OF EXPRESSION AND MEDIA DEVELOPMENT: GLOBAL REPORT 2017/2018 (UNESCO & University of Oxford 2018).

[20] *Id.* at 202.

[21] Allcott & Gentzkow *supra* note 4.

[22] Axel Gelfert, *Fake News: A Definition*, 38 INFORMAL LOGIC 84, 84-117 (2018).

[23] Madeleine de Cock Buning, OFF. OF THE EUR. UNION, A MULTI-DIMENSIONAL APPROACH TO DISINFORMATION: REPORT OF THE INDEPENDENT HIGH LEVEL GROUP ON FAKE NEWS AND ONLINE DISINFORMATION 11 (2018).

[24] Bertin Martens et al., EUR. COMMISSION, THE DIGITAL TRANSFORMATION OF NEWS MEDIA AND THE RISE OF DISINFORMATION AND FAKE NEWS (2018).

[25] Elle Hunt, *What is Fake News? How to Spot it and What you Can do to Stop It*, THE GUARDIAN (Dec. 17, 2016), https://www.theguardian.com/media/2016/dec/18/what-is-fake-news-pizzagate.

[26] *The Real Story of 'Fake News,'* MERRIAM-WEBSTER, https://www.merriam-webster.com/words-at-play/the-real-story-of-fake-news (last visited Jan. 21, 2023).

[27] Nic Newman et al., REUTERS INSTITUTE DIGITAL NEWS REPORT (2017). This definition adds two important dimensions (i) truthful information that misleads and (ii) truthful information that is disliked.

Multiple scholars, including those of constitutional law,[28] assert that author intent matters.[29] Satire, parody, and entertainment, without deceitful intent, do not constitute fake news.[30] Clarifying nomenclature distinguishes "misinformation," which can include "inadvertent sharing of false information" versus "disinformation," which refers to "deliberate creation and sharing of information known to be false".[31]

Others emphasize the role of the audience, further conditioning whether news is fake on the listener's belief as distinct from the author's intent.[32] They argue that without deception, fake news is but a work of fiction.[33] Interrogating subjective and unobservable factors such as author intent and listener belief are challenging but suggest the use of mechanism design to handle information asymmetry. "News you don't believe" is a murky colloquial[34] shortcut. If it is fake news that is disbelieved, then it does no damage and needs no intervention. If it is true news that is disbelieved, then it is not fake but could do damage and might need intervention, an irony that presents design challenges.

Put succinctly, the challenge of designing a fake news intervention based on the forgoing attributes appears to hinge on addressing veracity (falsifiability of news), velocity (speed and reach of news dissemination), and volume (amount and frequency of news production). It may also need to address author intent and listener belief. A leaner and different task, however, is presented in Section III.C.

B. Intervention Prospects

Fake news interventions to date have suffered from one or more of five fundamental problems. Technology solutions such as machine learning (ML) and artificial intelligence (AI) suffer from the *arms race* problem. If it is possible to train a filtering system to recognize fake news, it is also possible to train a competing system to recognize when content triggers recognition in order to avoid the triggers. The second problem is *discrediting the raters*. The actual accuracy of a rater matters little if those who propagate lies succeed in convincing their followers not to believe in the fairness of the rating system. Liars simply lie about the outcome, seeking to discredit the system that checks their distribution of lies. Governments, scientists, and other authorities have difficulty refuting misinformation when false claims

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[28] Cass R. Sunstein, LIARS: FALSEHOODS AND FREE SPEECH IN AN AGE OF DECEPTION (2021).
[29] Wardle & Derakshan, *supra* note 17.
[30] Under this view, Orson Welles' 1938 broadcast of a fake Martian invasion would not qualify as fake news, despite panic due to belief in its authenticity, as his purpose was entertainment not deceit.
[31] Claire Wardle, *Fake News. It's Complicated.*, MEDIUM (Feb. 16, 2017), https://medium.com/1st-draft/fake-news-its-complicated-d0f773766c79.
[32] Edson C. Tandoc Jr et al., *Defining "Fake News": A Typology of Scholarly Definitions*, 2 DIGIT. JOURNALISM 137 (2017).
[33] *See id.*
[34] Rasmus Kleis Nielsen & Lucas Graves, *"News You Don't Believe": Audience Perspectives on Fake News*, REUTERS INST. (2017), https://reutersinstitute. politics. ox. ac. uk/ourresearch/news-you-dont-believe-audience-perspectives-fake-news.

also impugn their intentions and capabilities.[35] The third problem is *misplaced responsibility*. A substantial fraction of interventions – crowdsourced fact checking, ML, AI, media literacy, accuracy nudges, tagging, truth chasers, demotion in the news feed – put the burden-of-proof on the platform or on the audience. The typical audience and the platform, however, have less information on content accuracy than the author, implying they are less equipped to police it. By analogy, these solutions place the burden of cleaning pollution costs on those who suffer pollution rather than on those who produce it. The fourth problem is *cost structure*.[36] Making up lies is cheaper than researching hard truths. This leads to a cost advantage for and oversupply of fiction over fact. The fifth problem is *conflict-of-interest*. Any decision body, whether public or private, runs the risk of decision bias when judging expression concerning itself. Effective solutions will need to address these deeper issues.

III. Truth, Falsity & The Nature of Harm

A. False Information Need Not Cause Harm

Truth is less clearcut that we might like to believe. Textbooks prior to 2006 published the existence of nine planets. In that year, the International Astronomical Union reclassified Pluto as a dwarf planet,[37] following the 2005 discovery of Eris[38] a celestial body with 27% more mass than Pluto. If the larger body was not a planet, then how could the smaller body be one? Texts prior to 1781 suggested there were only six planets.[39] Neither the six nor nine planet claim was true[40] yet the lives of few individuals changed in any meaningful way as new facts corrected old falsehoods.

One Asian fusion restaurant in Cambridge, Massachusetts used the slogan "Eat at Jae's and live forever!"[41] The National Enquirer, a publication whose circulation reached one million,

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[35] J. SCOTT BRENNAN ET AL., *Types, Sources, and Claims of COVID-19 Misinformation*, 1 (2020).

[36] *Id.*; Alcott & Hunt, *supra* note 21 ; Martens, et al., *supra* note 24.

[37] INT'L ASTRONOMICAL UNION, *Resolution B5: Definition of a Planet in Our Solar System*, https://www.iau.org/static/resolutions/Resolution_GA26-5-6.pdf (last visited Sept. 5, 2022).

[38] "Eris is the Greek god of discord and strife, a name which the discoverer Mike Brown found fitting in the light of the academic commotion that followed its discovery." INT'L ASTRONOMICAL UNION, *Pluto and the Developing Landscape of Our Solar System*, https://www.iau.org/public/themes/pluto/ (last visited Sept. 5, 2022).

[39] Uranus was discovered March 13, 1781 by William Herschel. Neptune was discovered Sept 23, 1846 by Urbain Le Verrier, Johann Gottfried Galle, and John Couch Adams. Pluto was discovered Feb 18, 1930 by Clyde Tombaugh. *History of Pluto*, LOWELL OBSERVATORY, https://lowell.edu/discover/history-of-pluto/ (last visited Jan 21, 2023).

[40] Logically, as long as the definition remains consistent, the number cannot be six or nine. If the definition includes dwarves, then the set of planets includes previously undiscovered bodies Eris and Ceres for a set of at least eleven. If one excludes dwarf planets, then the set includes only eight.

[41] Amy Graves & Wesley Morris, *At the New Jae's Café, Diners Can Still Pad Thai Like It's 1990*, BOS. GLOBE (2005) http://archive.boston.com/dining/globe_review/1087/. Jae's discontinued using the slogan after several locations closed.

published a cover story that actress Rita Hayworth had returned from the dead.[42] These claims are provably false. Neither claim, even in total aggregate, has produced social costs rising to a level that would require regulatory oversight.

Even the belief in false news need not cause harm.[43] One might believe that vaccines do not work yet still take them in compliance with the law or one might believe the world is flat yet still take cruises that sail the globe. Those beliefs had no ill effect. Only when one acts on beliefs that cause harm does fake news lead to a social interest in curbing the harm. Then also, restricting actions can occur without revising beliefs; they represent separate points of intervention with different ethical and efficiency considerations.

In fact, lies can create immense social value. Undercover officers lie to thwart drug cartels and terrorist organizations.[44] Members of the underground railroad, who lied to bounty hunters after passage of the Fugitive Slave Acts of 1793 and 1850, faced fines and imprisonment but helped thousands of refugee slaves escape to freedom.[45] During the Holocaust, villagers of the town Vivarais-Lignon lied to Gestapo officers about the presence of Jewish refugees, risking their own execution, yet they helped desperate people evade Nazi concentration camps.[46] Collectively, such lies saved hundreds if not thousands of people from incarceration, immiseration, and death.

Falsity alone is not a metric that can determine the need for intervention.

B. True Information *Can* Cause Harm

During World War I, the English language North China Daily News printed allegations that a German factory was rendering human corpses into fats to produce nitroglycerine and lubricants.[47] This story was false and intended to gain allies in the war. During World War II, however, Nazi Propagandist Joseph Goebbels used the truth about these stories to discredit other true stories of German war crimes against the Jews. In effect, he used truthful news to discredit truthful news.[48]

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[42] *I'm Back From the Dead – For Two Years I was a Zombie*, NAT'L ENQUIRER, https://www.pulpinternational.com/pulp/entry/National-Enquirer-with-Rita-Hayworth-cover-from-1963.html (last visited Jan 21, 2023).

[43] YUVAL NOAH HARARI, 21 LESSONS FOR THE 21ST CENTURY 233 (2018). ("As long as we all believe in the same fictions, we all obey the same laws, and can thereby cooperate effectively.").

[44] Chen, A. K. (2022). Investigative Deceptions Across Social Context. *U Denver Legal Studies Research Paper*, (06). xxx

[45] *Fugitive Slave Acts*, HISTORY (Feb. 11, 2020), https://www.history.com/topics/black-history/fugitive-slave-acts.

[46] Maggie Paxson, *What We Can Learn About Being Good from a Village That Saved Thousands During the Holocaust*, TIME (Sept. 19, 2019), https://time.com/5680342/french-village-rescued-jews/.

[47] *The Corpse Factory and the Birth of Fake News*, BBC (Feb. 17, 2018), https://www.bbc.co.uk/news/entertainment-arts-38995205.

[48] RANDAL MARLIN, PROPAGANDA AND THE ETHICS OF PERSUASION (2013).

"To tell a truth with ill intent beats all the lies you can invent." – William Blake 1919

In 2018, a US federal court banned the free dissemination of blueprints for 3D printed guns. If the blueprints had been inaccurate or fake, they could not have been used to print working guns. The judge banned them on the basis that harm to the private defendant's *First Amendment* rights "are dwarfed" by the harm States might incur if anyone could print a gun.[49] Criminals could defeat security scanners. Printed plastics would facilitate terrorist hijackings. Persons legally barred from gun ownership, due to prior conviction, restraining order, or mental condition could summarily obtain them.

In 2016, Russian troll accounts Blacktivist and DrConservaMom used social media to broadcast true stories of white officers shooting black men and of school shootings. Their messages used true information, tinged with political spin, to suppress black votes in neighborhoods favoring democrats and to animate gun rights voters in neighborhoods favoring republicans. While it is logical to cull such messages on the premise of foreign interference in sovereign elections,[50] it violates free speech to do so when propagated by domestic citizens themselves. Citizen accounts assumed this role in the 2020 US elections.[51]

Among ideologues and propagandists, it is common practice to take two truths and add them to create a lie. A famous celebrity who did die and who did receive a vaccine are used to show the danger of vaccination although the vaccine had nothing to do with his cause of death.[52] Extending their toolkit, they may take three truths and subtract one, again to create a lie. A famous politician who did die of covid despite receiving a prophylactic vaccine is held up as evidence of vaccine inefficacy, omitting the fact that blood cancer suppressed his immune system.[53] Antivaxxers routinely cite a small piece of evidence from legitimate research, remove context, and grossly exaggerate it.[54]

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[49] Washington v. U.S. Dept. of State, 318 F.Supp.3d 1247, 1264 (2018).

[50] Annie Palmer, *Facebook Removed 3.2 Billion Fake Accounts Between April and September, More than Twice as Many as Last Year,* https://www.cnbc.com/2019/11/13/facebook-removed-3point2-billion-fake-accounts-between-apr-and-sept.html (Nov. 13, 2019).

[51] *See* SINAN ARAL, THE HYPE MACHINE: HOW SOCIAL MEDIA DISRUPTS OUR ELECTIONS, OUR ECONOMY, AND OUR HEALTH—AND HOW WE MUST ADAPT (2020); *see also* Alyza Sebenius, *Fake Social Media Accounts Seen as Threat in November Election*, https://www.bloomberg.com/news/articles/2020-09-01/social-media-impersonators-seen-as-threat-in-upcoming-elections (Sept. 1, 2020).

[52] Shannon Bond, *Just 12 People Are Behind Most Vaccine Hoaxes on Social Media*, NPR (MAY 14, 2021), https://www.npr.org/2021/05/13/996570855/disinformation-dozen-test-facebooks-twitters-ability-to-curb-vaccine-hoaxes.

[53] Keenan Chen, *Vaccine Skeptics Seize on Death of Colin Powell to Spread Misinformation*, FIRST DRAFT (Oct. 22, 2021) https://firstdraftnews.org/articles/vaccine-skeptics-seize-on-death-of-colin-powell-to-spread-misinformation/.

[54] *Id.*

Veracity alone is not a metric that can determine the need for intervention.

Much false information causes no harm. Much true information does. Indeed, "truth" is a poor candidate for the operation of law. One cannot own truth. One cannot be liable for truth. One cannot be dispossessed of truth. If one cannot assign property rights in or liability for an entitlement, legal mechanisms focused there will have little force.[55] Rather, mechanism design should focus on *decisions* for which one can be liable and whose ill-gotten gains may be alienable.

Of forgoing truthful examples that do cause harm, each represents either a decision error or a negative *externality*. In the case of Goebbel's misuse of true information, the harm was not to Goebbels or readers of his fake news but to the Jews who suffered historic atrocities. In the case of working instructions for printed guns, the harm is not to the author of blueprints or the consumer who prints them but to innocent victims who are shot using them. In the case of voter suppression, it is not just the citizens who don't vote but all other citizens who become governed by a different choice of candidate. In the case of antivaccination exaggerations, not only do the unvaccinated err based on true but incomplete information but also the failure of herd immunity and economic costs accrue to the entire community.

C. Problem Statement

Fake news constitutes a form of information pollution from which there extend potentially large externalities. The need to address externalities as well as consequences of veracity leads to the following problem definition.

> *Externalities cause market failures; market failures require intervention, but the First Amendment forbids government intervention in speech. Thus, attempts by courts to use the "marketplace of ideas" to sort things out will fail. Markets do not self-correct market failures.*

This explains why the fake news problem is hard. Pre Internet, a limited number of news outlets, who remained liable for falsehoods, produced a tolerable level of pollution. Post social media, where anyone can be amplified and the amplifiers cannot be held accountable for falsehoods,[56] the pollution level is intolerable. With no way to internalize externalities, our current design for a marketplace of ideas prevents correction. To address this problem, this Essay then takes up the following task:

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[55] Guido Calabresi, & A. Douglas Melamed, *Property Rules, Liability Rules, and Inalienability: One View of the Catherdal*, 85 HARV. L. REV. 1089 (1972).
[56] Wu *supra* note 3. See also §230 of the 1996 Communications Decency Act.

> *Design mechanisms to clear communications channels of information that causes harm through decision errors or negative externalities propagated at scale.*[57]

Causes of decision error include false information and true but incomplete information. Causes of negative externalities are involuntary costs imposed on third parties by the voluntary communications of any self-interested parties.

This problem statement and associated task differ from prior interpretations in two important ways. First, focusing on decision error rather than truth renders the task easier. One can be liable for decision error or dispossessed of ill-gotten decision gains whereas one cannot be liable for truth or dispossessed of truth. It also separates the problem from intent, which may or may not be discernable. Intent might matter at a penalty phase but is rarely available at the decision to disseminate phase. This problem statement also implicates half-truths that deceive listeners into choosing differently than how they would have chosen under full information. At the same time, it exonerates parody and irony that cause no decision error. Indeed, using irony, the opposite of one's true meaning, is a linguistic device to create emphasis, which can cause the listener to choose more carefully, reducing the chances of error.[58] It also exonerates misstatements that are factually incorrect but not of a magnitude that change a decision. Second, this problem statement adds a missing component, harm to third parties. A solution that only envisions harm to the listener fundamentally misses the negative externalities that harm others. Neglecting externalities is one of the profound mistakes made in present day free speech jurisprudence.[59] It is also one of the reasons that platforms, despite seeking to maintain healthy interactions on-platform, have caused devastating harms off-platform.[60] In his CEO testimony before Congress describing Facebook's interaction with Cambridge Analytica, Zuckerberg stated "We did not take a broad enough view of our responsibility."[61] Cambridge Analytica promised games and surveys to Facebook users but in exchange harvested not only all their individual data but also their friends' data, a textbook example of a third-party externality. Friends did not grant

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[57] This task statement has a historical parallel in the form of Shannon's channel coding theorem, which seeks to establish whether an information set generated at a source can be communicated without error across a noisy channel to a destination. The answer is yes so long as information volume does not exceed a capacity determined by the noise level. C.E. Shannon, *A Mathematical Theory of Communication*, 27 BELL SYS. TECH. J. 279 (1948). In human terms, the comparable notion to channel capacity is "bounded rationality." *See* the Paradox of Individual Liberty in Marshall Van Alstyne, *Free Speech, Platforms and the Fake News Problem* (Dec. 31, 2021) (available at http://ssrn.com/abstract=3997980).

[58] I thank Joel West for this observation.

[59] *See* Van Alstyne, *supra* note 57.

[60] Paul Mozur, *A Genocide Incited on Facebook, with Posts from Myanmar's Military*, N.Y. TIMES (Oct. 15, 2018), https://www.nytimes.com/2018/10/15/technology/myanmar-facebook-genocide.html.

[61] *Facebook, Social Media Privacy, and the Use and Abuse of Data, Senate Committee on the Judiciary, Senate Committee on Commerce, Science, and Transportation*, 115th Cong. 1 (2018) (Testimony of Mark Zuckerberg, Chairman and Chief Executive Officer, Facebook), https://web.archive.org/web/20180418004240/https:/www.judiciary.senate.gov/imo/media/doc/04-10-18 Zuckerberg Testimony.pdf.

permission for data harvesting and were unaware of the breach.[62] The spillover consequence for others is precisely the issue of third-party harm.

With this problem definition, solution concepts then target different points of leverage. The first effort seeks to allow third parties harmed by misinformation the means to undo the damage they suffer. Lack of transparency has hindered the market for counter speech. Enabling this market is business model compatible. The second effort seeks to have those producing externalities internalize their harms. No longer granted free disposal of their pollution, platforms do not find this business model compatible. The third effort seeks to develop decentralized interventions such that no one party – not government, not private enterprise, not influential individuals – exercises control over verification processes. The grant of new rights in a decentralized market is business model compatible. Taking the free market of ideas as metaphor, the task becomes market repair.[63]

IV. Solutions Derived from Choice Architecture

The presence of an externality means that knowledge of the transaction is divorced from knowledge of the harm. Information sets do not overlap; decision and effect are disjoint. A framework for addressing the problem then admits one of two solutions. The first is a governance mechanism that moves knowledge of the harm to the party with knowledge of the transaction. Facebook, for example, could learn of damage that its amplification causes. The second is a governance mechanism that moves knowledge about the transaction to the party with knowledge of the harm. People affected by Facebook's amplification, for example, could learn details of harm spreading.

The first option is inferior for at least three reasons. (i) Concentrating all information at the center creates a platform of large power and little oversight. Moving off-platform information on-platform has the potential to increase information asymmetry between members of society and the platform increasing risk of widespread exploitation. (ii) Moving information from off-platform to on-platform does not align incentives. Since the party with knowledge of the transaction is *not* the party suffering harm, the central platform has little incentive to change behavior to improve social welfare. (iii) The near infinite variety of potential externalities ensures that certain forms of damage are likely to be missed. Pulling all possible information with all possible externalities onto the platform is a daunting, if not impossible, task. In effect, the reasons that Facebook "did not take a broad enough view of [its] responsibility" include both that it is technically infeasible and incentive incompatible. Moreover, were it possible to succeed, the outcome would not be desirable.

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[62] Carole Cadwalladr, *'I Created Steve Bannon's Psychological Warfare Tool': Meet the Data War Whistleblower*, THE GUARDIAN (Mar. 18, 2018), https://www.theguardian.com/news/2018/mar/17/data-war-whistleblower-christopher-wylie-faceook-nix-bannon-trump.

[63] Information economics should be uniquely well-suited for addressing such questions yet has played little role in legal scholarship to date. *See* Daniel J. Hemel, *Economic Perspectives on Free Speech,* U. CHI. COASE-SANDOR WORKING PAPER SERIES IN L. & ECON. (2019)*;* THE OXFORD HANDBOOK OF FREEDOM OF SPEECH 118-136 (Frederick Schauer & Adrienne Stone eds., 2019).

The second option is superior for reasons that invert the logic above. (i) Moving information that is on-platform to third parties off-platform decentralizes power and reduces information asymmetry, reducing risk of exploitation. (ii) Parties that suffer harm obtain information on the causes of harm. Incentives align, giving those with the desire to act the information needed to act. Welfare naturally improves. (iii) Public exposure, moving on-platform information off-platform, facilitates parallel search by diverse members of a society. This is decentralized rather than centralized governance. The chances for uncovering the nature of harm improve. At the same time, decentralization fosters a marketplace of information where different ideas compete. This offers a first solution.

1) Equivalence of Counter Speech Access: Pair knowledge of externality with access to means of resolution – grant "equivalence of access" to listeners, not simply transparency into speakers, sufficient to allow reversal of decisions.

This is not merely a matter of transparency or maintaining records for a period of years and making the nature of an ad, advertiser, and contact information available. It must also provide access to those who saw a claim and to those where it spread. This was infeasible in print and broadcast media. In terms of a decision rights doctrine, this solution provides people means of influencing third party decisions that affect them, to *undo* the damage, not merely allow for its discovery. Information must be actionable not merely knowable. To the extent that counter-speech rebuts lies,[64] these speaker decision rights provide remedy by providing access to targets equivalent to those obtained by the parties causing injury. On balance, this solution should be business model compatible as it simply allows one partisan to use the same tools as any other partisan.

Media platforms promote demagogues in the name of newsworthiness.[65] Resolving information asymmetry between those doing the harm and the locus of harm suggests a first intervention. Media platforms should provide access to recipients of misinformation. Access should include, for example, not only people reached by a claim but also those contacts to whom such claims were propagated. Access goes beyond current transparency requirements that record who purchased an ad, its content, and release dates. Transparency only lets affected parties learn that damage has occurred. It does not provide means of undoing the damage. Limited disclosure only provides information sufficient to reach the perpetrator, not the means to undo harm by enabling counter messaging. By contrast equivalence of access allows an injured party to seek redress not only by holding the perpetrator accountable, which transparency provides, but also by updating facts and narratives among recipients, which access equivalence provides.

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[64] Eugene Volokh, *When Are Lies Constitutionally Protected?*, REASON (Jun. 18, 2020). https://reason.com/volokh/2022/07/18/when-are-lies-constitutionally-protected/.

[65] Whistleblower testimony revealed a secret list exempting politicians and cultural icons from rules against spreading misinformation. Jeff Horowitz, *Facebook Documents Reveal Secret Elite Exempt From Its Rules,* WALL ST. J. Sept. 14, 2021.

Transparency laws that simply record the nature of a communication – its sponsor and its content – do not per se address decision errors or externalities. If a crime were committed, such laws provide only for recording the event or compensating it without undoing the damage. Transparency laws allow people to discover how they were injured and to hold the speaker accountable. They do not, however, enable the injured party to reach those members of the market for ideas where the false idea has taken root. Transparency might allow one to sue a candidate for a lie placed prior to an election but access can allow a competing candidate to *undo* the lie before that election. If awareness of an externality is insufficient to correct it, then, in order to fix it, access to means must accompany knowledge of ends. Media platforms must therefore record and provide access to recipients of ads received directly as well as those who received them indirectly via sharing and propagation.

Market access need not affect privacy of individual recipients. Neither their identities nor their contact information need be disclosed. Instead, the platform simply mediates access.

Importantly, media platforms should find provision of equivalent market access to be business model compatible. They need not arbitrate truth. They simply sell ads or enable reach with access to all parties equally. In effect, "Equivalence of Access" on social media resembles the existing though disused FCC "Fairness Doctrine" that requires broadcasters to provide equal market access to present their case if requested. "Equivalence of Access" is both distinct from and weaker than the "right of reply" that required free placement of rebuttals on behalf of citizens disparaged in broadcast editorials. The Supreme Court upheld this FCC rule in *Red Lion*[66] on the basis of limited bandwidth.[67] A person who felt attacked by a radio station had few ways to reply and launching a competing radio station under government granted licensing was impractical. A legally binding free response then balanced broadcasters' editorial rights, speakers' reach rights, and audience diversity needs. By contrast, Equivalence of Access requires neither equal time nor free access but only an equal channel to audience targets at published and prevailing rates. As with access to patent pools, terms should be Fair Reasonable And Non-Discriminatory (FRAND). It balances the media platform's interest in protecting its assets and the social concern with fair markets for ideas. Government plays no role in adjudicating content and, if involved, only decides whether market access is fair.

The Supreme Court invalidated a Florida state right to reply in *Miami Herald Publishing v. Tornillo* in the case of newspapers.[68] In contrast to broadcast, print is not licensed by government. While Florida's rule did not prevent editors from saying what they wished, "it exact[ed] a penalty on the basis of the content" and because finances are limited, "editors

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[66] Red Lion Broadcasting Co. v. FCC, 395 U.S. 367, 401 (1969)

[67] Stanley Ingber, *The Marketplace of Ideas: A Legitimizing Myth*, DUKE L. J. 1, 58 (1984). As explanation, the court wrote, "It is the purpose of the *First Amendment* to preserve an uninhibited marketplace of ideas in which truth will ultimately prevail, rather than to countenance monopolization of that market, whether it be by the government itself or a private licensee…" Red Lion, 395 U.S. at 390. For further analysis of monopoly and the marketplace of ideas, see Van Alstyne, *supra* note 57, at 31-36.

[68] 418 U.S. 241 (1974).

may conclude the safe course is to avoid controversy."[69] By contrast, Equivalence of Access boosts revenues and, if anything, invites crosstalk among political opponents.[70] Ironically, Equivalence of Access, paid at FRAND rates, has the opposite effect of the court's concern for an economic burden on the press. Rather it introduces the moral hazard that a media platform could invite or offer critique so as to prompt those affected to buy a response. Controversy boosts engagement. It is certainly business model compatible. The truth of each perspective could then surely enter the market.

While counter speech access gives those suffering harms a means to undo them, it provides no incentive for those supplying harms to not produce them. Platforms amplify controversy because lies spread "farther, faster, deeper and more broadly than truth in all categories of information."[71] This creates a perverse incentive to ignore externalities in pursuit of profit.

At present, we know of only two methods to force those producing externalities to internalize their harms. The first, proposed by Arthur Pigou, levies a tax proportional to the damage in order that marginal private cost rises to marginal social cost. The producer then internalizes harms rather than shifting them to society. The second, proposed by Ronald Coase, creates property rights in the externality and uses markets to trade and price it.[72] This raises the cost of otherwise free disposal, while shifting the burden of harm to whomever can bear it most cheaply. We develop each option in turn.

V. Solutions Derived from Pigou

Nobel laureate Paul Romer has endorsed a variant of the Pigouvian tax intended to focus on the business model rather than the externality per se.[73] Legal scholars Netanel[74] and Salib[75] have likewise proposed an excise tax on digital platforms. Applying a Pigouvian tax to digital ad sales offers at least three advantages over alternative interventions. First, an ad tax directly alters the business model by favoring ad-free subscription revenue over ad-based third-party revenue. Subscription revenue need not require user tracking. Also, above a minimum participation threshold, subscription revenue does not create an incentive to artificially boost engagement. User privacy could improve even as fake news driven demand declined. Second, a progressive tax, with higher costs for larger firms, could favor entrepreneurial startup. Larger firms created by smaller firm mergers would face larger ad

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[69] Id. at 257.

[70] For a stronger solution to this problem that enables the piercing of filter bubbles see Solutions Derived from Coase in Van Alstyne, *supra* note 57, at 41-46.

[71] Soroush Vosoughi, Deb Roy, & Sinan Aral, *The Spread of True and False News Online*, 359 SCI. 1146 (Mar. 9, 2018).

[72] Ronald Coase, *The Problem of Social Cost*, 3 J. L. & ECON. (1960); Herbert J. Hovenkamp, *The Coase Theorem and Arthur Cecil Pigou*, 51 ARIZ. L. REV. 633 (2009).

[73] Paul Romer, *A Tax That Could Fix Big Tech*, N.Y. TIMES (May 7, 2019), https://www.nytimes.com/2019/05/06/opinion/tax-facebook-google.html.

[74] Neil Weinstock Netanel, *Mandating Digital Platform Support for Quality Journalism*, 34 Harv. J. L. & Tech. 474 (2021).

[75] Peter N. Salib, *The Pigouvian Constitution*, 88 *U. CHI. L. REV.*, 1081 (2021)

tax bills. Relative to breakup, a progressive ad tax simultaneously solves a market concentration problem normally solved through competition law but does so more effectively. Two half size firms could produce more damage than one full size firm if the principal effect of breakup were to cause each smaller firm to compete more fiercely for user engagement. Not so for the progressive ad tax. Third, an ad tax is neither content nor consequence based, which avoids most free speech concerns. Because it does not involve operational oversight, it reduces risk of regulatory capture that can arise from oversight.

2) Pigouvian Solution A: Apply a progressive tax to media advertising

Romer's version of a Pigouvian tax has two shortcomings. First, the primary means by which all platforms create value is by consummating matches.[76] They pair people with friends, news, apps, search results, rides, movies, products, and destinations. Effective matching requires tracking. Purported privacy benefits will not fully materialize although subscription revenues do align the interests of user and payer in a way that ad revenues do not. Second, one of its greatest strengths is also its greatest weakness. The damage targeted by an ad tax is unhealthy levels of engagement not misinformation per se. By avoiding content issues, the ad tax divorces the levy from the externality. A private subscription service could host antivaxx disinformation, conspiracy theories, and false election narratives but pay no tax, whereas a clean ad-driven service, free of fake news, could pay a heavy tax. For Pigou's solution to work, the penalty must scale with the externality. In the context of factory pollution, a tax on effluent volume encourages the factory to shift to less harmful technology but only if the volume corresponds to the harm. A factory could reduce total effluent volume by increasing pollution concentration (and actual damage) to reduce the tax. By contrast, if the tax applies to the pollutant specifically, the factory shifts to less polluting technology.

This insight offers a means of reforming a progressive ad tax to improve efficacy: tie the levy to the concentration of harms produced in news effluent. This functions exactly in the manner of taxing the concentration of harmful effluent in factory production. Yet, a tax on amplified falsehood spillovers acts differently than a tax on specific speech. Testing a statistically valid sample solves three fundamental problems, one of scale, one of accuracy, and one of law. First, one need not certify every message; rather a certification authority need only validate a random sample to achieve any confidence level desired. A doctor testing for cholesterol does not test every drop of blood but only a statistically valid sample. Sampling could even apply to closed chat rooms without violating the privacy of the individuals involved. Second, in a rigorous mathematical sense, a flow rate or aggregation of signals provides a constantly updating Bayesian credibility score. Based on the Central Limit Theorem, larger samples cause estimates of any parameter to converge closer to truth as samples accumulate. This advantage is enormous as it deals even with mixed stories that blend truth with lies. Individual message context becomes far less important because errors only need to fall within an acceptable confidence interval, which can be determined by

[76] Geoffrey G. Parker, Marshall Van Alstyne, & Sangeet Paul Choudary, PLATFORM REVOLUTION: HOW NETWORKED MARKETS ARE TRANSFORMING THE ECONOMY AND HOW TO MAKE THEM WORK FOR YOU (2016).

sample size. An overarching news credibility score characterizes fitness with respect to the whole message population and not simply a single event. Parties on the left and right might disagree on which messages are true yet agree more readily on the flow rate of truth. Given sufficient statistical samples, consistent deviation from an average score can indicate bias in a specific critic as easily as bias in a specific critique. A third benefit is that the boundaries can scale easily with maturity. A large mature platform might face an effluent rate of .1% while a startup could face one of 1% or 10%. Holding platforms accountable need not disadvantage entrants that lack prior experience. Lastly, a fourth benefit is that a tax on concentration of harm offers a practical means to weaken the liability protections of Section 230 while retaining its broader benefits. The binary choice between either complete liability immunity or accountability for all individual messages is too coarse. Platforms can reasonably object that policing content of 500M individual messages daily is not practical. Societies can reasonably object that policing disinformation that causes political insurrections, deaths from infections, and genocidal riots needs to be practical, or the platform should not operate. A reasonable balance – one that can adapt to different societies – is to hold platforms accountable for a specific fraction of effluent they amplify. Tax the preponderance of dysfunctional information as distinct from specific instances of disinformation. Measuring a fraction of effluent can exhibit practical scale, adapt to law, and converge to truth.

3) Pigouvian Solution B: Tax the platform in proportion to measurable harms. Make it progressive based on size and maturity.
4) Section 230 Reform: Separate liability for users' original posts and platforms subsequent amplification, generously protecting the former but reverse amplifying the latter.
5) Section 230 Reform: Determine liability for amplification on a statistical rather than individual message basis

How might a Pigouvian mechanism operate in practice? The pollution problem is one of joint choices: one party chooses to post and another chooses to amplify. How can we hold amplifiers accountable, without chilling speech, in a manner that is also fair? A simple solution illustrates.

Suppose that a scientist creates an invention of economic value 1 and a venture capitalist builds a business around it, increasing its value 1000 times. The venture capitalist might wish to claim a (1000)/(1+1000) = .999 share of business value but that is obviously unfair. Nobel laureate Lloyd Shapley provides the fair answer[77] by asking who is indispensable to each value increment. For the first increment, the scientist is indispensable and receives 100% of 1. For the next increment of 1000, the scientist and venture capitalist are *both* indispensable and so each receives 50% of 1000. The shares are (1 + 500) / (1+1000) = .5005 to the scientist and (500)/(1+1000) = .4995 to venture capitalist. Regardless of

[77] Lloyd Shapley, *A Value for n-person Games*, 2 *CONTRIBUTIONS TO THE THEORY OF GAMES* 307-17 (Kuhn & Tucker eds. 1953).

whether the outcome is innovation benefit or misinformation harm, the same fair formula holds.

If a platform were charged 49.95% for its share of damage, which is less than half the total social cost, its liability would fairly cover its share. If this makes amplifying lies unprofitable, the platform stops amplifying, and because damage was *jointly* produced, the full 1000 units fall away, leaving only the original 1. Free speech is protected. There is no chill even though it causes 1 unit of harm, while all 1000 units of extra harm disappear.

A practical solution is then straightforward and can even avoid costly assessments of harm. If platforms pay a bit more than the ad revenues they generate from amplifying lies, then that business becomes unprofitable, and amplified damage disappears. As a free speech issue, although the true social cost might be 1000, a platform could still amplify misinformation aligned with its views for what is little more than the price of its own ads. Only now, those ad revenues would go to society. Use of the *Central Limit Theorem* together with the price of a platform's own ads allows for a mechanism that is fair, predictable, and low cost.

Conservatives and liberals have critiqued Section 230 but for seemingly opposite reasons.[78] Those on the right fear platform firms exhibit bias against conservative voices. They wish to condition platforms' immunity from prosecution on not censoring posts. By contrast, those on the left protest that platforms promote disinformation for profit because they cannot be held accountable. They wish to condition immunity on good faith efforts to remove harmful content.[79] Splitting the original post from platform amplification can split the difference and is consistent with expert advice.[80] Reforming Section 230, an original post can receive greater free speech protection than private firms currently offer. Reform can also ensure that platforms must treat all speakers equally in contrast to their biased practice of exempting influential speakers from content moderation despite their stated policies.[81] Yet, statistical measures of platform amplification of terrorist recruiting,[82] self-harm content,[83] regulated goods,[84] medical misinformation,[85] or foreign election interference[86] can be used to hold platforms accountable on a fair and reasonable basis.

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[78] Tim Wu, *Liberals and Conservatives are Both Totally Wrong about Platform Immunity*, Medium (Dec. 3, 2020). https://superwuster.medium.com/liberals-and-conservatives-are-both-totally-wrong-about-section-230-11faacc4b117.
[79] Id.
[80] ASPEN INSTITUTE, Commission on Information Disorder – Final Report (Nov. 15, 2021). https://www.aspeninstitute.org/publications/commission-on-information-disorder-final-report/
[81] Ingber, *supra* note 67.
[82] *Gonzalez v. Google LLC* (Docket 21–1333); *Twitter, Inc. v. Taamneh* (Docket 21-1496).
[83] META TRANSPARENCY CTR., https://transparency.fb.com/policies/community-standards/suicide-self-injury/ (last visited Feb. 26, 2023).
[84] For information on drugs, explosives, hazardous materials etc., see META TRANSPARENCY CTR.. https://transparency.fb.com/policies/community-standards/regulated-goods/ (last visited Feb. 26, 2023).
[85] Bond, *supra* note 52.
[86] Id.

Used as "sin taxes" for such problems as pollution, alcohol and cigarettes, Pigou's tools can adjust to illegal content by culture. Germans might target hate speech and Nazi propaganda.[87] Saudi Arabia might target violations of Sharia.[88] These would be illegitimate U.S. uses, yet Pigou's tools could pass strict scrutiny by targeting incitement to imminent violence, child pornography, stolen intellectual property, advertising illegal drugs, and foreign election interference. Regardless of culture, Russia's Vladimir Putin has no right to interfere in Saudi, German, or American elections so when platforms amplify such content, governments have an interest in removing it.

Though powerful and effective, Pigou's tools remain problematic because they use a central authority to judge pollution levels and levy fines, yet a central authority is anathema for judging or fining speech concerning itself.[89] Coase's tools, however, are market based.[90] The task then is to design systems of rights such that decentralized institutions can internalize negative externalities.

VI. Solutions Derived from Coase

An idea dating to Adam Smith's invisible hand[91] is that successful economies depend on voluntary exchange. The decentralized choices of buyers and suppliers make efficient use of private information to allocate resources. Economists such as von Hayek[92] give this as the reason market economies create more wealth than centrally planned economies. In mathematical terms, the *First Fundamental Welfare Theorem* of economics proves this idea formally: producers and consumers, each acting in their own self-interest, yield a socially efficient Pareto outcome.[93] Thus individuals should make their own decentralized choices rather than have a central government chose for them.

By analogy, speaker and listener choices to produce and consume content might match those of suppliers and buyers to produce and consume goods. Who has better private information on production costs than the producer? Who has better private information on

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[87] Dan Glaun, *Germany's Laws on Hate Speech, Nazi Propaganda & Holocaust Denial: An Explainer*, FRONTLINE (July 1, 2021), https://www.pbs.org/wgbh/frontline/article/germanys-laws-antisemitic-hate-speech-nazi-propaganda-holocaust-denial/.

[88] *See* Articles 1 and 39 of the Basic Law of Governance of Saudi Arabia.

[89] JOHN S. MILL, ON LIBERTY (Batoche Books Kitchener 2001) (1869).

[90] Coase, *supra* note 72.

[91] ADAM SMITH, AN INQUIRY INTO THE NATURE AND CAUSES OF THE WEALTH OF NATIONS (Seligman, 1901) (1776).

[92] Friedrich A. von Hayek, (1945). The Use of Knowledge in Society. *American Economic Review*, 35(4), 519-530. Compare, for example, the United States, England, and Japan to North Korea, East Germany, and the former Soviet Union. See Kramer, L. "Market Economy versus Command Economy" https://www.investopedia.com/ask/answers/100314/whats-difference-between-market-economy-and-command-economy.asp. Von Hayek received the 1974 Nobel prize in economics for his work showing a decentralized market system with free competition and pricing uses diverse actors' knowledge better than central planning.

[93] MAS-COLELL, A., WHINSTON, M. D., & GREEN, J. R. (1995). MICROECONOMIC THEORY (Vol. 1). New York: Oxford university press.

whether speech is a lie than the speaker? It probably is not the government. If this metaphor holds, a marketplace of ideas might discard centralized speech decisions in favor of decentralized citizen decisions. Indeed, U.S. courts have repeatedly held that "the best test of truth is the power of the thought to get itself accepted in the competition of the market."[94]

The problem with this metaphor, as well as Holmes' argument, is that the *First Fundamental Welfare Theorem* fails in the context of information asymmetry and of externality. Lies exhibit both, which destroys the social efficiency properties of voluntary exchange. Market failures born of information asymmetry and of externality require intervention to avoid "information rent seeking" and to "internalize externalities."

Nobel prize winning economists have supplied tools to address *both* these market failures and restore a higher level of efficiency to voluntary exchange. The information asymmetry solution is the more straightforward of the two because it involves only the transacting parties i.e., buyers and sellers in product markets or listeners and speakers in idea markets. Signals and screens, warrants and bonds, can address the adverse selection and moral hazard problems of misrepresenting one condition for another.[95] The externality solution is harder because it involves not only the listener and speaker but also third parties hurt by listener and speaker choices. Hence, government action has seemed necessary to protect these third parties or to make them whole. Compared to the anarchy of no oversight at all, content moderation has seemed necessary for free expression to flourish.[96]

Centralized solutions are Pigouvian solutions. Censorship, viewed differently, is a quota on certain kinds of poisonous speech in the manner that industrial pollution controls are a quota on certain kinds of poisonous effluent. A central bureaucrat decides whether effluent levels are tolerable and punishes the malefactor to force internalization of the externality.

Coase, however, supplies a second set of tools.[97] Harmful externalities arise from a missing market in untraded harms. If a society can identify new property rights in these harms and assign liability to the lowest cost avoider, negotiations between buyer and seller – or listener and speaker – can achieve an outcome that is at least as welfare enhancing as any decision by government. This solution works because it simultaneously reduces the harms, by forcing their internalization, and because it removes the less-informed central authority from negotiations. From a free speech perspective, this is obviously *First Amendment*

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[94] Abrams v. United States, 250 U.S. 616 (1919) (Holmes, J., dissenting).

[95] Michael Spence, *Job Market Signaling*, Q. J. OF ECON. (1978); PETER DIAMOND & MICHAEL ROTHSCHILD, UNCERTAINTY IN ECONOMICS (1978); George A. Akerlof, *The Market for "Lemons": Quality Uncertainty and the Market Mechanism*, UNCERTAINTY IN ECON. (1978); Joseph E. Stiglitz & Andrew Weiss, *Credit Rationing in Markets with Imperfect Information*, 7 AM. ECON. REV. 393-410 (1981). These authors received the 2001 Nobel prize in Economics for insights into economics of information.

[96] Mike Masnick, *Why Moderating Content Actually Does More to Support the Principles of Free Speech*, TECHDIRT (Mar 30, 2022). https://www.techdirt.com/2022/03/30/why-moderating-content-actually-does-more-to-support-the-principles-of-free-speech/

[97] Coase, *supra* note 72. Coase received the 1991 Nobel prize in economics both for his insights into transaction costs and the nature of the firm and how *property rights* affect the functioning of the economy.

compatible. Whether the central authority is a government, or a social media platform, does not matter. Only the source and destination in decentralized exchange are involved.

A contribution of this Essay is to propose such a system of rights. The solution is an *expansion* in the rights of listeners and of speakers in a manner that balances their conflicting interests, the listener's right to be free of speech harms and the speaker's right to be heard.

Let a *listener's right* grant individuals the entitlement to seek information from trusted or preferred sources *of their own choosing*. Information seeking qualifies as a fundamental human capability.[98] Logic decentralizes this choice to the listener, rejecting the tendency of government and of industry to arrogate this capability to themselves on the basis that gubernatorial and industrial self-interest are not identical to that of the individual.[99] Granting individuals a *listener's right* – the ability to choose from competing sources that would aggregate, filter, emphasize, and vouchsafe content – supports a market in content moderation. Conservative, liberal, scientific, artistic, and comedic moderators each competing to serve individual choice might then provide "diverse and antagonistic" sources from which truth could emerge. This individual right to choose asserts an equal right *not* to choose and, with it, the right to hear and *not* to hear.[100] As liberating as this right may be, its free exercise can lead to balkanization[101] and filter bubbles[102] as listeners may choose to hear only what they chose to believe.

At the same time, let a *speaker's right* entitle individuals to such free expression as grants them the ability to influence any decision that affects them also in a manner *of their own choosing*. Like information seeking, information dissemination also qualifies as a fundamental human capability.[103] As before, logic decentralizes this *speaker's right* to individuals, even when they knowingly authorize others to speak on their behalf, to preserve one's right to safeguard one's own interests. The rights to seek and to disseminate information should be inalienable, otherwise grievous injustice could occur. To influence decisions entails being heard. If one disagrees with another speaker, one needs the right to counter disagreeable speech. As liberating as this right may be, its free exercise can lead to half-truths and complete lies as people seek advantage for themselves at the expense of others.

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[98] Martha C. Nussbaum, *Capabilities as Fundamental Entitlements: Sen and Social Justice*, 9 Feminist Econ. 33 (2007).

[99] For a discussion of *Decision Rights Doctrine* testing these listener and speaker rights in the context of Supreme Court cases, see Van Alstyne, *supra* note 57, at 22-31.

[100] Expressed differently, this listener's right further implies a right to privacy to the extent that the right to hear and *not* to hear implies a right to be left alone.

[101] Van Alstyne, M., & Brynjolfsson, E. (2005). *Global village or cyber-balkans? Modeling and measuring the integration of electronic communities*. Management science, 51(6), 851-868.

[102] Pariser, E. (2011). The filter bubble: How the new personalized web is changing what we read and how we think. Penguin.

[103] Nussbaum, *supra* note 98.

A conflict of rights necessarily follows. How does the right to be heard offset the right not to hear? If everyone speaks, can anyone hear? If some can be heard, which claims are worth hearing? If some claims cause harm, whose rights are worth curbing?

Without answers to these questions, the marketplace of ideas descends into a market of lemons. It is cacophony and chaos. Struggling for an audience, those with valuable ideas may exit the market. Not bearing costs, those telling lies may disproportionately enter. A decentralized market might operate through individual choice, but if and only if those choices balance speaker and listener rights, provide indicia of quality, and internalize decision harms and externalities.

Bridging information economics and externality economics, we expand and rebalance speaker and listener rights as follows.

In this new market, speakers are free to express their claims as facts or opinions, but facts attach a new entitlement. A speaker gains the right to have factual claims heard, even over listener or moderator objection, by warranting that content is valid, meaning it is not per se illegal and it is not materially false.[104] This time-limited warrant is any resource placed at risk, be it social capital (e.g. followers or reputation points) or financial capital (e.g. currency or resources), posted in advance and, like hazard insurance, set to cover possible harms. A listener or any third-party objecting to warranted facts is free to challenge its claims. To dispute a claim, a challenger pays a modest fee to cover its adjudication cost. Peer juries might ensure both legitimacy and accuracy.[105] The resource placed at risk is a reward to the challenger, when the claim is judged false, but is returned to the speaker when the claim is judged true.[106] If a claim goes unchallenged, all resources return to the speaker. Opinions cannot be warranted for validity and so cannot be imposed on others' newsfeeds without their (or their moderator's) consent. All warrants are voluntary. No speaker need warrant any content but then listeners need not listen, and moderators need neither carry it nor amplify it. In legal terms, this entitlement can be viewed as defining a listener right to be free of hearing false claims, protected by a liability rule. Alternatively, it is a speaker right to have true claims heard, subject to a liability rule.[107] Independent decisions made without warrants merely reflect the status quo *ex-ante* where speakers, listeners, and third parties bear their own costs of not being heard, of hearing false claims, and of suffering externalities respectively.

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[104] E.g. the Pope endorsed my candidate, vaccines contain microchips, the election is Wednesday not Tuesday. Claims must also be *verifiable*. Attaching a warrant to a negative claim must still meet the burden of proof.

[105] Low-cost social media peer juries have been shown to reliably identify misinformation. *See* Jennifer Allen, Antonio A. Arechar, Gordon Pennycook, & David Rand, *Scaling up Fact-Checking Using the Wisdom of Crowds,* 7 SCI. ADVANCES 1 (2021).

[106] Courts have not enforced prove-me-wrong offers without consideration. See Hemel, *supra* note 63. Here, the quid pro quo of a guarantee in exchange for having to hear or carry content establishes consideration.

[107] Calabresi & Melamed, *supra* note 55.

7) Solution derived from Akerlof, Spence & Stiglitz: Create a market for truth. Let any party wishing to signal the truth of factual claims place a resource at risk as warrant those claims are valid (i.e. not illegal and not false).
8) Solution derived from Coase: Choose the expected value of the resource at risk to reflect the expected cost of externalities. These social harms, which scale with audience size, set the "price of lies." In exchange for accepting liability at the price of lies, allow a speaker's right to be heard to override a listener's right not to hear.

This signaling and trading mechanism exhibits several important properties. (i) If a claim is false, then its author knowing it be false will not want to guarantee its veracity. (ii) If a claim is true, however, authors can voluntarily "signal" integrity because honest guarantees are costless.[108] This addresses the *cost structure* problem that fake news is cheaper to produce than honest news. (iii) Together, these properties yield a separating equilibrium based on authors' private knowledge, distinguishing misinformation from authoritative information. Listeners learn whom to believe and whom not to believe. Determination of human behavior, not technological determination of content, avoids the *arms race* problem. (iv) Initial burden for deciding truth rests with the author rather than a platform or an uninformed listener. The author is free to phrase a claim in any manner most suited to withstanding challenge. This is more socially efficient and addresses the *misplaced responsibility* problem. (v) Truth markets protect whistleblowers. To expose a crime, a person warrants a claim anonymously and only needs to provide evidence to withstand a challenge. By contrast, a reluctance or inability to warrant suggests false accusers cannot provide evidence. The separating equilibrium depends on the signal value of the warrant, not the person's identity. (vi) It *internalizes harmful externalities*. Any third-party hurt by a false claim is free to dispute that claim. A successful challenge moves the social cost from the party experiencing pollution to the party sourcing that pollution. This reduces pollution. (vii) Decentralized crowdsource detection of falsehoods scales. Members of the crowd with private knowledge of falsity are motivated to unmask false claims by the reward. *Ex-post* verification of individual claims is also easier than *ex-ante* screening of all claims. (viii) The challenge fee discourages false challenges. It also covers the costs of adjudicating a challenge, rewarding jurors, so the mechanism is financially self-sustaining. (ix) Adjudication *censors no content*. The best supporting evidence, pro and con, becomes a public record. Verdicts achieve not only contestation but also resolution among "diverse and antagonistic sources."[109] (x) Impartial hearings before random juries are hard for ideologues to discredit relative to standing bodies whose verdicts, though true, they dislike. This addresses the *discrediting the rater* problem. (xi) Outcomes are consistent with *First Amendment* jurisprudence. All choices are voluntary. The escrowed warrant is an independent signal that simply allows honest speakers to distinguish themselves. (xii) The

[108] Michael Spence, *Job Market Signaling*, Q. J. OF ECON. (1978); PETER DIAMOND & MICHAEL ROTHSCHILD, UNCERTAINTY IN ECONOMICS (1978); George A. Akerlof, *The Market for "Lemons": Quality Uncertainty and the Market Mechanism*, UNCERTAINTY IN ECON. (1978); Joseph E. Stiglitz & Andrew Weiss, *Credit Rationing in Markets with Imperfect Information*, 7 AM. ECON. REV. 393-410 (1981). These authors received the 2001 Nobel prize in Economics for insights into economics of information.
[109] Assoc. Press v. United States, 326 U.S. 1, 20 (1945).

mechanism establishes a principled basis not just for speech but also for reach. A speaker may reach any audience of larger size, even one that objects to a message, in exchange for accepting more liability for false claims. It simultaneously protects audiences from hearing false claims and gives them recourse for falsehoods they had not wished to hear. (xiii) The entire mechanism is decentralized, and market based. By design, *no central authority* exists. Like the *Fundamental Welfare Theorem* of economics, outcomes depend on speaker-listener choices, not those of a central planner. This avoids the *conflict-of-interest* problem of having any central authority judge truth.[110]

Importantly, a market for truth works even when the social cost of damage is not known in advance. A fossil fuels company might warrant a claim that human activity does not cause global warming[111] because its profits vastly exceed the current lie price. If a firm repeatedly loses challenges and continues to pay the lie price, then the price of that lie can rise until it stops causing harm. Escalation provides a search process that can force liars to internalize the true social cost of their negative externalities even when that cost began as an unknown. Given uncertainty, social efficiency sets the lie price at the expected level of harm[112] yet the starting point can adjust to social taste. Cultures that value freer expression can start the lie price lower while cultures that value information integrity can start the lie price higher. Analogous to car accidents, repeated infractions should raise the liability in any culture. Here, rising liability dissuades liars from repeated lying, finds an efficient price, and limits entrenching the lie.

Interestingly, establishing a "market for truth" is also business model compatible. It removes responsibility for adjudicating truth from the social media platform, returning this to society, yet it enables the platform to participate in the advertising and escrow markets. Conditional on building the institutional infrastructure necessary to support a truth market, a Coasian solution combined with information economics is economically sustainable. Such a market, analogous to that for carbon trading, could address the misinformation problem.

A. Socially Efficient Media Liability

To further illustrate their power, truth market warrants can help address the longstanding media liability and speaker reach problems in paid advertising. Inconsistencies in print, broadcast, and internet media liability illustrate legislative advertising policies that emerged based on technologies of their times rather than any deeper principle. Each medium faces different rules. Under current law,[113] print publishers face the strictest ad liability. They are free to use their discretion to accept or reject ads but if they print an ad containing false

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[110] Collectively, these address each of the intervention challenges identified in Section II.B.

[111] Jeffrey Pierre & Scott Neuman, *How Decades of Disinformation About Fossil Fuels Halted U.S. Climate Policy*, NPR (Oct 27, 2021), https://www.npr.org/2021/10/27/1047583610/once-again-the-u-s-has-failed-to-take-sweeping-climate-action-heres-why.

[112] Louis Kaplow & Steven Shavell, *Property Rules Versus Liability Rules: An Economic Analysis*, 109 HARV. L. REV. 713 (1995).

[113] Fed. Trade Comm'n Act of 1914, 15 U.S.C §53(d).

content, they can be held liable.[114] Broadcast media face a "no censorship, no liability" rule for political candidates.[115] They are not free to reject candidates' ads but in exchange for this lack of freedom, broadcasters also face no liability for candidates' false claims. Interestingly, broadcaster liability changes for non-candidate advertising. They are free to reject ads from non-candidates but if they accept ads with false claims, then they can be held liable.[116] The third and most privileged category are the Internet platforms who have both discretion over what to accept and immunity from liability regardless of editorial decisions.[117] These inconsistencies in liability for paid advertising apply independent of the fact that all three categories – print, broadcast, and platform – depend heavily on ads for revenue. Across all three, political interests that are not candidates can be denied access to a given audience if their message, however true, conflicts with editorial policies of the medium. Discourse diversity suffers. Is there means to balance interests of all parties – polities, media, and citizens?

When transaction costs are high, liability for pollution should rest with the lowest cost avoider.[118] Markets can trade their way to efficiency *unless* these costs are high. With low negotiating costs, as between a city and nearby factory spewing foul air, assigning property rights to either party matters little if one can buy rights from the other. Voluntarily trading payments for moving the factory, compensating the citizens, or changing the technology will create more social value than edicts set by government fiat, especially if either side has hidden costs or benefits that a government cannot observe. Assigning property rights *does* matter if negotiations must take place between each citizen and the factory. Then assigning liability to the factory, the lower cost avoider, reduces total transaction costs, while still enabling trade. Media that spews misinformation is a lower cost avoider than is each citizen who may or may not know ad veracity and may or may not have resources to check it. Should the medium, which is paid to disseminate an ad, check that ad's claims one time or should 100,000 citizens, who are not paid, check that ad 100,000 times? Social efficiency demands the former, else it asks 100,000 people to wear gas masks or purify their water rather than ask one factory to clean its effluent.

Between an ad's author and an ad medium – print, broadcast, or internet – who should bear liability? How can a society balance a speaker's interest in reach, a medium's interest in editorial autonomy, and an audience interest in hearing diversity? Truth market warrants offer a simple and principled solution. On the principled basis of Coase, Calabresi and Melamed, let liability vest initially with the medium. But, if an ad's author will warrant a claim, the medium must accept and disseminate the message. In exchange, the author

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[114] Mark Sableman, *When Politi-Fact Veers into Politi-Fiction, Broadcast and Internet Advertising Rules Diverge Sharply*, J.D. SUPRA (Nov. 27, 2019), https://www.jdsupra.com/legalnews/when-politi-fact-veers-into-politi-46971/.
[115] Communications Act of 1934, 47 U.S.C. § 151 et. seq.
[116] Sableman, *supra* note 114.
[117] Communications Decency Act of 1996, 47 U.S.C. § 230. The Communications Decency Act was struck down by the Supreme Court in *Reno v. ACLU*. Only § 230 survived. *See* Reno v. ACLU, 521 U.S. 844 (1997).
[118] Coase, *supra* note 72; Calabresi & Melamed, *supra* note 55; Kaplow & Shavell, *supra* note 112.

assumes liability absolving the medium. The medium is free to disseminate any message it likes but retains liability for any ad the author does not warrant. Political interests may reach any audience merely by assuming liability for their claims. Media are free to accept or reject any unwarranted ad. Media retain liability for ads the author will not warrant and from which they choose to profit. The party with editorial authority retains liability. The transactions burden is not shifted to citizens, placing solution of the pollution problem onto the lowest cost avoider. Diversity of discourse improves as speakers willing to warrant their claims can reach any audience. Truthfulness rises. Polarization falls.

VII. Tests & Objections

The complexity of intervening in news streams means that balancing different objectives inevitably breaks some constraint. This Section seeks to address the most common objections to interventions in any market for free speech. These four are among the most common.

1. Platforms do not produce the content they propagate.

If social media platforms do not author the content that cause harm, why should having them internalize these costs be more efficient than having content creators bear these costs? Section 230 grants them immunity on this basis. A more robust analysis provides two answers based on governance and transaction costs. First, social media platforms already internalize the positive externalities of social networks; they need only internalize the negative externalities as well. Social media platforms are built on and derive their power from network effects. Their purpose is to foster connections.[119] The contacts and activity of one user benefit other users. These are externalities and when they are positive the platform already encourages them and profits from them by interposing itself and monetizing ads between connections. When harms occur on-platform, as in the case of harassment or fraudulent products, the platform already addresses them. Platforms only need motivation to take actions they already take yet must do so for harms that occur off-platform in addition to those that occur on-platform.

Although social media platforms do not author the content they dispense, any claim they exert no influence over members is disingenuous. They actively engage in orchestration. They are the governments of their ecosystems with authority to regulate speech,[120] participation, prices, competition, and intellectual property within their regimes.[121] The

[119] Facebook's original mission statement was "Making the world more open and connected." Josh Constine, FACEBOOK CHANGES MISSION STATEMENT TO 'BRING THE WORLD CLOSER TOGETHER' TECHCRUNCH (June 22, 2017), https://techcrunch.com/2017/06/22/bring-the-world-closer-together/.
[120] Kate Klonick, *The new governors: The people, rules, and processes governing online speech.* 1598 Harv. L. Rev. 131 (2017).
[121] GEOFFREY G. PARKER, MARSHALL W. VAN ALSTYNE, & SANGEET PAUL CHOUDARY, PLATFORM REVOLUTION 158 (2016); JEAN TIROLE, ECONOMICS FOR THE COMMON GOOD (2017).

venture capitalists who invest in platforms, not merely the economists, have stated as much.[122] When the citizens of one country suffer the pollution of another, the government of the former might reasonably negotiate with the government of the latter, especially when polluters in the latter are invisible to citizens of the former. Although neither government itself produced the pollution, lax rules in the source country are at least partially responsible for pollution in the harmed country. In this case, the costs of harmed individuals bargaining with each polluting firm greatly exceed those of collective bargaining. This argues for negotiating with the government of the polluting country as the highest leverage point of intervention.

Second, platforms do, in fact, represent the nexus of lowest transaction costs.[123] Social media platforms, unlike one-way broadcast media, orchestrate the activities of their users. In order to facilitate membership and engagement, they reduce friction on participation and production. They provide tools for creation, tools for consumption, and feedback on impact. In fact, no party has greater visibility than social media platforms into the nature of misinformation transactions. Without information supplied by the platforms, not even the authors themselves know who has shared or who has read their campaigns. Transaction cost economics weigh in favor of intervening at the point of lowest cost and greatest transparency, in this case, the point of the platform. Social media platforms represent the point of greatest leverage.

2. How can this or any mechanism decide what is true?

The effort to ascertain truth has two approaches, one practical and one theoretical. In practice, courts routinely grapple with the question of whether a claim is true to a given level of certainty. The weakest standard for a burden of proof is a *preponderance of evidence*, meaning that a claim is more likely true than not.[124] On balance, its probability exceeds 50%. The middling standard is *clear and convincing* evidence, meaning a claim is highly and substantially more likely true than false. The trier of fact must be convinced it is highly probable.[125] The highest standard is *beyond a reasonable doubt*, meaning that evidence is so compelling as to leave no other explanation.[126] Alternate hypotheses have been ruled out. Similarly, the laws around "duty of care" for product liability are particularly vague and differ by state in the same manner that free speech laws differ by country. And yet, as a practical matter, we deal with them. We apply local context and change the threshold for certainty according to the severity of the decision.

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[122] Brad Burnham, *Web Services as Governments*, UNION SQUARE VENTURES (June 10, 2010), https://www.usv.com/blog/web-services-as-governments.
[123] Michael Munger, *Coase and the Sharing Economy,* in FOREVER CONTEMPORARY: THE ECONOMICS OF RONALD COASE 187 (Cento Veljanovski ed. 2015).
[124] People v. Miller, 171 Cal. 649, 652 (1916).
[125] Colorado v. New Mexico, 467 U.S. 310 (1984).
[126] Thomas Gallanis, *Reasonable Doubt and the History of the Criminal Trial*, 76 U. CHI. L. REV. 941 (2009).

As a theoretical matter, we cannot know absolute truth. This objection simply re-asks Hilbert's "*Entscheidungsproblem*" in a new context. The Church-Turing thesis tells us that certain statements cannot be proven true or false. A more precise statement of Hilbert's decision problem, grounded in logic and philosophy, is as follows. Given a system of claims, is it possible to definitively prove the collected assertions are true? The answer, in general, is no. Posed in 1928, this hard question was not answered until 1936, when Alonso Church and Alan Turing independently developed methods to prove that an infinity of claims are undecidable. Modern computer theory provides an interesting clarification. If statements are made at one end of a communication channel, can one be certain that identical statements are received at the other end of that channel? In effect, data corruption – literally false news introduced into the channel – can be repaired using error correcting methods but only up to a point. Shannon's Channel Coding Theorem, which forms the basis of all modern communications, proves that arbitrarily small error in communicating a fact is not achievable. Error correction is only possible up to a fixed and finite boundary. It is impossible past that boundary.

Thus, the question presupposes an answer that cannot be given. This objection is used to dispatch any approach that cannot solve the problem, which is an unfair critique because *no* approach can solve the problem. We can only know truth to a given number of bits. Interestingly, this comports well with the practical solution of accepting a claim as true "beyond a reasonable doubt."

3. Edge cases between true and false invalidate the mechanism. Whatever the boundary conditions, it is always possible to split a boundary with careful wording.

The existence of an edge case is not a legitimate challenge to any governance mechanism generally, let alone a fair news mechanism specifically. There does not exist any useful mechanism for which there do not exist edge cases between true and false. The decision criterion "Always Guilty" has no edge cases but also no practical application, as does its opposite "Always Innocent". Even our most cherished and most absolute rights have edge cases. Is the right of free speech absolute? We admit slander, libel, and incitement to violence as exceptions. Is the right to life absolute? We admit self-defense as an exception. The existence of edge cases can be used to exclude *every* mechanism, which leaves only the null mechanism, thus it fails as a legitimate test. The proper test of regulation is not perfection but rather how it performs compared to its best alternative.[127] Rather, the test should be whether one mechanism adjudicates edge cases better than the alternative. Importantly, the alternative is not the null set of no mechanism at all. In our case, the alternative is the present mechanism being used by social media platforms and, judged in terms of efficacy and absence of bias, those leave much to be desired. The proper challenge is therefore to articulate the alternative mechanism and show why it does better than the

[127] Cary Coglianese & Evan Mendelson, *Meta-Regulation and Self-Regulation*, OXFORD HANDBOOK OF REGULATION, 148, 161 (Cave, Baldwin & Lodge ed. 2010)

proposal under consideration. Across a weighted sum of false positives and false negatives, which rule achieves more social value? Admittedly, this is a high bar. The best challenge is a superior mechanism design. The best and most challenging objection is thus an act of creation and not merely an act of rejection.

4. Who gets to decide?

Reasonable people will disagree and those who dislike a decision will seek to discredit the decision maker. Unless adjudication is indisputably impartial, partisans will not accept results.[128]

Conservatives may reject a decision whose outcome favors a liberal view. Liberals may reject a verdict whose outcome favors a conservative view. This objection raises separate issues of reconciling opposing views and of decision legitimacy. On the issue of reconciling conflicting views, there are three reasons why *requiring* agreement is ill advised.

i. People do not universally wish to be convinced of a position nor do they grant third parties the moral authority to convince them. They often reject data that disagrees with their identity or world view or position in life. "It is difficult to get a man to understand something when his salary depends on him not understanding it."[129] More carefully, one may reasonably ask, what gives the mechanism designer the moral authority to assert the righteousness or truthfulness of the mechanism's verdict?[130] Absent such authority, perhaps the empowered view should shift its position to the disempowered view.
ii. Mechanisms that *require* agreement cause moral hazard. If partisans know they will be bought out, with resources needed to convince them, they can exaggerate their claimed protests and supposed beliefs. These beliefs are not themselves verifiable. The social cost to overcome this mechanism-induced moral hazard could be in excess of the value of the verdict, producing social waste. The alternative, coercion, risks reaching the wrong conclusion simply by placing the power of coercion with one or another party.
iii. The most compelling reason, however, that requiring agreement is not a valid test is an artifact, again, of the *Entscheidungsproblem*: knowing absolute truth is absolutely impossible. If one *unbiased* party cannot know or even communicate certain truth, it is pointless to require multiple *biased* parties to agree on that truth. Universal agreement is an impossible standard.

If reconciliation is infeasible, then in what sense might a verdict be legitimate? The solution is one we recognize in other contexts as procedural fairness.[131] Partisans must agree *ex-ante*

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[128] This objection arose in a March 4, 2021 conversation with the misinformation team at Facebook.
[129] UPTON SINCLAIR, *I, CANDIDATE FOR GOVERNOR* (1994).
[130] John Stuart Mill, *On Liberty,* in A SELECTION OF HIS WORKS (1966).
[131] Kees Van den Bos, Henk A. M. Wilke, & E. Allan Lind, *When Do We Need Procedural Fairness? The Role of Trust in Authority* 75 J. OF PERSONALITY & SOC. PSYCH. 1449 (1998).

to the method for deciding what's true, then commit to abide by the impartially administered verdict.

Thus, to operate a market for truth, we can rely on established administrative practices that we already use for trust and legitimacy. Taking our own government as precedent, consider a design where we split fake news oversight into legislative, judicial, and executive offices. A legislative body gets to define "fake news." Despite their differences, liberal and conservative media might be able to agree on a working definition independent of specific cases. A judicial body gets to decide whether a specific case represents an instance of fake news according to this definition. Fact-checking organizations or juries of peers might play this role only now they must judge according to the definition provided by the legislative body. Jurors do not get to use their own individual definitions. Finally, the executive branch enforces these definitions and decisions. Social media platforms like Facebook and Twitter can play this role but they decide neither the definitions nor the outcomes of challenges. Such an institutional structure could operate as a self-governing trade association or independent standards body. If self-governance fails to emerge, government can apply such pressure as needed to implement "meta-governance," inducing regulatory targets to develop their own self-regulatory practices.[133] By dividing the branches of fake news governance, we recreate an institution where no branch judges truth as applied to itself, and no branch has an economic incentive to bias its behavior to get rich. The divided process should therefore be free of conflict of interest, less biased due to random sampling, and by design more legitimate.

The broader design point, however, is to *decentralize* intervention in the marketplace of ideas so that no one party gets to decide – not government, not private enterprise, and not powerful individuals. Truth Market design, as presented in Section VI, offers one such mechanism.

VIII. Conclusions

Societies cannot function that cannot agree on facts. Is the planet warming or not? Do pandemic vaccines work or not? Is the president in office legitimate or not? Twenty-first century fake news has pushed the boundaries of our institutions past the boundaries of our twentieth-century frameworks.[134]

The goal of this Essay has been to reshape our understanding of the problem in order that we might use different tools to shape better answers to clearer questions. One step in this process is to shift the framing from a focus on truth to a focus on decision errors and externalities. Truth cannot be owned nor can those who have it be alienated from it. Truth makes a poor candidate for the operation of law. By contrast, one can own the

[132] Nat'l Fed'n of Indep. Bus. Et. Al. v. Sebelius, 567 U.S. 519 (2012).
[133] Coglianese & Mendelson, supra note 127.
[134] Bak-Coleman et al., *supra* note 13, at 3.

consequences of decisions and be dispossessed of their gains. One can be liable for externalities. Focusing on truth per se overlooks the fundamental nature of why the free speech versus fake news problem is hard. Free expression has externalities. Externalities cause market failures. Market failures require intervention, yet the *First Amendment* forbids government intervention in speech. This paradox makes the problem hard when the practice of jurisprudence has been to turn the resolution of truth over to the market which has no current means to clear itself of externalities. The trick then, is to design interventions that do not involve government. The harder trick is to design decentralized mechanisms that do not simply transfer the authority we deny government to any other institution capable of molding others' expression to its own self-interest. Current law grants platforms authority over user speech but absolves them from abuses of that authority.[135]

Information economics provides us with a new set of tools. Within this set, the first design options draw from choice architecture. Better designs reverse amplification of dysfunctional information and seek to place the burden for disseminating falsehoods on the author rather than the medium or the reader. They also go beyond traditional forms of transparency in order that parties harmed by misinformation can do more than learn of the harm; they can act to reverse it. Information access together with reach aligns incentives and provides the means for injured parties to counter speech that causes harm.

Recognizing misinformation as a pollution problem summons a second set of tools – those of Pigou and those of Coase – to address externalities. These grant the power to force those who disseminate damage to internalize that damage, overturning the incentive to propagate falsehoods for private gain at social cost. These also give us means to assign liability to platforms for their share of harms. We can thus revise Section 230 in ways that protect speech more than platforms do, avoid harms jointly created by users and platforms, and scale with the volume of content to any level of certainty we desire. These tools can reform the business models of the twenty-first century information giants in the manner they reformed the pollution habits of the twentieth century industrial giants.

Combining the tools of externality and information economics makes possible signaling and screening in a Market for Truth. To create this market, this Essay proposes a new pair of rights to enable trade in a missing market of harms. One entitlement defines a listener right to be free of hearing false claims, protected by a liability rule. The other entitlement defines a speaker right to have true claims heard, subject to a liability rule. A primary advantage of this market is that, by design, no central agency judges content. Neither states nor institutions exercise concentrated power. Rather, informed authors, who know they are honest, can credibly signal integrity of their claims. Uninformed readers can then distinguish dishonest authors who are unwilling to warrant their dishonest claims. The hope is that decentralized choices of individual information seeking and dissemination support a truth market in the manner that individual choices of production and consumption support a

[135] Communications Decency Act of 1996, 47 U.S.C. § 230.

price market.[136] A secondary advantage is a change in cost structure. To date, no other mechanism has addressed the integrity handicap that producing fake news costs less than producing true news.[137] These tools can also pierce polarizing filter bubbles while reconciling the inconsistent laws in paid advertising that have afflicted print versus broadcast versus internet media. Such laws arose from times and places past and have long needed reform.

[136] In the ideal, this would achieve a social optimum in the manner of the *First Fundamental Welfare Theorem* of economics. See *supra* note 93.

[137] With the introduction of accessible and convincing chatbots, this gap may only grow in the coming years. Tiffany Hsu & Stuart A. Thompson, *Disinformation Researchers Raise Alarms About A.I. Chatbots*, N.Y. TIMES (Feb. 13, 2023) (detailing how researchers used ChatGPT to produce convincing text that boosted fake news).